

# Synthesizing Staged Updates between Verified Controllers

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Verified old and new controllers do not supply a safe transition when component replacement and requirement activation require different schedules. Fine-Grained Dynamic Update Controller Synthesis (FG-DUCS) compiles local transfers, requirement lifetimes and initial obligations into a policy between supplied controllers. Winning means safe completion from every reachable old entry under every permitted outcome, enforcing declared lifetimes and preserving the selected new controller’s continuation. We prove contract/game correspondence and specialize established lazy reachability to return a policy and progress rank or a losing-region certificate. Complete examples show merging commands can prevent or enable completion. Within 1,200 s, our MTSA implementation decides 26/27 adapted contracts, versus 17 for eager successors, 24 for update-first exploration and 14 for full construction. Update-first is sometimes faster. Constructed families exhibit granularity effects; inherited inputs exhibit none. Commands and handover require uncontrollable quiescence; continuous service and elapsed-time bounds are not guaranteed.

CCS Concepts: • **Software and its engineering** → **Software evolution**.

Additional Key Words and Phrases: dynamic software updating, discrete-event controller synthesis, supervisory control, on-the-fly synthesis

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## 1 Introduction

A verified controller governs one version. Updating requires reaching another while component states and active requirements change [3, 6, 14, 37]. The endpoints do not supply this intervening policy. We generate it while preserving the supplied new controller’s behavior after handover.

Components and requirements pose different scheduling choices. Empty-station guards can require moving a workpiece before replacing each station. Audit coverage may require overlap between old and new requirements, whereas exclusive roles require a gap.

The policy must cover different old states and every environment outcome while retaining duties such as inspection before departure. A fixed sequence may fail under branching; safety alone permits an update to remain unfinished forever.

*FG-DUCS compiles local choices into a policy for safe finite handover.* The contract fixes permitted transfers, requirement lifetimes and endpoint obligations; synthesis chooses how to complete the update (Table 2). Modeling and transfer code remain the modeler’s responsibility; effort reduction is unmeasured.

DUCS [32, 33] and GR(1) updating [2] already generate update or bridge control. FG-DUCS instead fixes both controllers and generates the intervening game from local transfers, independent requirement boundaries and initializers (Section 2.3). Its contribution is this construction and its guarantees, not greater general expressiveness or a new reachability algorithm.

Our contributions are:

**C1: A local contract and its meaning.** We define the generated game, prove local-history correspondence and selected-endpoint preservation, and give an invariant-initialization criterion (Sections 3 and 4.1).

**C2: Boundaries of update granularity.** FG-DUCS examples and merging obstructions explain when separate choices are necessary and when they prevent completion (Sections 2 and 5).

**C3: A policy with checkable evidence.** Sound and complete lazy synthesis returns a completing policy and rank or a losing-region certificate (Section 6).

**C4: Implementation evidence.** An MTSA implementation is checked against finite reference games and evaluated on 27 adapted contracts; constructed comparisons test the granularity mechanisms separately (Section 7). All unfavorable and unresolved outcomes are retained.

The target is a finite maintenance interval. Work may pause; commands and handover require global uncontrollable quiescence. Interval requirements cover duties between old-stop and new-start. Neither continuous service nor elapsed-time bounds follow; an environment that forever avoids quiescence prevents completion.

Our public replication package [4], prepared for this paper, includes the cited Technical Appendix A–P and Supplement S1–S5 as Parts I and II of `supplementary_material.pdf`. Data Availability gives versioned links.

## 2 From a Complete Contract to a Completing Policy

Component granularity chooses what is replaced together; requirement granularity chooses which guarantees start or end together. Two contracts separate these choices.

### 2.1 Cell: Complete Local Input and Whole Policy

Consider a constructed two-station cell with exactly one workpiece [22, 33]. Both transfers require an empty station.  $A$  keeps its behavior; new  $B$  requires inspection before departure, while the old rule forbids inspection. Ordinary events move or inspect the workpiece. New-start before old-stop prevents an unprotected gap; a pending inspection waits for old-stop. Inspection can repeat safely forever.

Table 1 specifies every component, endpoint and monitor. The interval requires exactly one workpiece, which plant transitions preserve. All labels are controllable; new-start before old-stop is the only dependency. Transfers are  $g_A = g_B = \{(e, e)\}$ , where  $e$  means empty.

Table 1. Complete Cell components, controllers and monitors.  $o/n$  denote old/new versions;  $h/e$  mean holding/empty. Parentheses mark initial states.  $m_A, m_B$  move to  $A, B$ ;  $j$  inspects at  $B$ .

| Object               | States (initial)       | Ordinary transitions                                                                                           |
|----------------------|------------------------|----------------------------------------------------------------------------------------------------------------|
| $A^o, A^n$           | $h, e (h)$             | $h \xrightarrow{m_B} e, e \xrightarrow{m_A} h$                                                                 |
| $B^o$                | $h, e (e)$             | $e \xrightarrow{m_B} h, h \xrightarrow{m_A} e$                                                                 |
| $B^n$                | $h, e (e)$             | $e \xrightarrow{m_B} h, h \xrightarrow{m_A} e, h \xrightarrow{j} h$                                            |
| $C^o$                | $c_0 (c_0)$            | $c_0 \xrightarrow{m_A, m_B} c_0$                                                                               |
| $C^n$                | $c_0, c_1 (c_0)$       | $c_0 \xrightarrow{m_A, j} c_0, c_0 \xrightarrow{m_B} c_1, c_1 \xrightarrow{j} c_0, c_1 \xrightarrow{m_B} c_1$  |
| $t_{\text{one}}$     | $A, B, \text{err} (A)$ | $A \xrightarrow{m_B} B, B \xrightarrow{m_A} A, A \xrightarrow{m_A} \text{err}, B \xrightarrow{m_B} \text{err}$ |
| $t_{\text{old}}$     | $k, \text{err} (k)$    | $k \xrightarrow{j} \text{err}$                                                                                 |
| $t_{\text{inspect}}$ | $c, d, \text{err} (c)$ | $c \xrightarrow{m_B} d, d \xrightarrow{j} c, d \xrightarrow{m_A} \text{err}$                                   |

$k, c, d$  mean old-safe, inspection-clear and inspection-pending;  $c_0, c_1$  are distinct controller states.

Component/controller alphabets are exactly their displayed labels; unlisted edges are absent. Unlisted non-error monitor edges self-loop, and `err` absorbs every label. Each comma-separated label denotes an edge.

Endpoint tuples list  $A, B$ , controller and its requirement monitor. The old initial tuple is  $x_A = (h, e, c_0, k)$  and its other reachable tuple is  $x_B = (e, h, c_0, k)$ ; both are entries. The new initial tuple is  $z_A = (h, e, c_0, c)$ ; its other reachable tuples are  $y_B = (e, h, c_1, d)$  and  $z_B = (e, h, c_0, c)$ . Only  $\{z_A, z_B\}$  are loadable targets: inspection must be clear at handover. The interval initializer records the holder in either all-old tuple. The new initializer admits all eight version-tagged one-workpiece tuples, installing  $d$  if  $B$  holds and  $c$  otherwise. These data specify the contract without an update sequence.

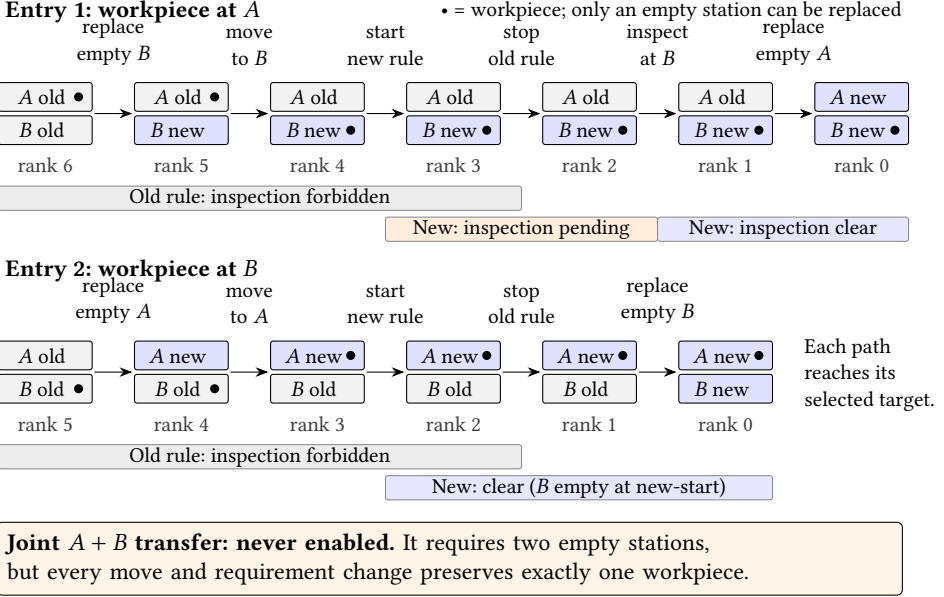
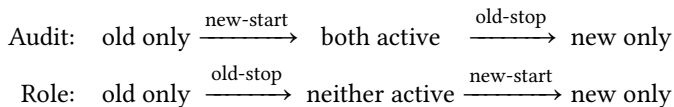


Fig. 1. A complete local policy from both old entries. Bands mark active rules and new-rule inspection memory; new-start precedes old-stop. Each arrow has one action and one outcome, decreasing rank toward its selected new target. Joint transfer merges these local commands with endpoints and requirements unchanged; this is not a DUCS encoding.

*The generated policy.* Figure 1's 13 states and 11 edges cover both entries and targets. Each displayed event has one outcome; other controls are disabled. Both paths move the workpiece to enable empty-station replacements. The first also clears inspection debt before handover. Every arrow decreases rank from six or five to zero, excluding safe but endless inspection. A joint transfer always requires both stations empty and is impossible. Technical Appendix B and S1 give the paths and closure. Section 2.3 compares the supplied and generated structures through existing update interfaces.

## 2.2 Audit and Role Lifetimes

Policy(2) has two old/new requirement pairs: audit and role. Four bits record active rules  $a_o, a_n, r_o, r_n$ . The interval invariant  $(a_o \vee a_n) \wedge \neg(r_o \wedge r_n)$  requires opposing boundary orders:



The new audit monitor starts unlogged and requires a log before service. Each endpoint alternates its own log and service events and permits only its version’s access role. With an identity component transfer, the saved Lazy policy first performs service if the old audit is logged, starts new audit, stops old audit and role, transfers, and starts the new role. Service is optional; this policy reaches a compatible new endpoint from both old states, with rank at most six.

If the starts are merged and the stops are merged, the first start overlaps roles and the first stop removes audit coverage. Ordinary events and transfer preserve the four activity bits, so neither old entry has a completing merged policy. Technical Appendix C gives the complete finite input, policy and losing closure. This compares merged and separate FG-DUCS requirement boundaries.

## 2.3 Existing Update Interfaces

Table 2. Supplied inputs and generated results: DUCS [33, Defs. 4.1–4.3, Sec. 5], GR(1) updating [2, Def. 3.1, Sec. 4], and FG-DUCS.

| Interface      | Supplied inputs                                                                                                | Generated structure and successful result                                                                | Post-update control                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| DUCS           | Old control; old/new environments $E, E'$ and goals; global transfer relation $R$ ; transition requirement $T$ | Update environment and goal; controller $C'$ and map $H$ covering every old-controller state             | Synthesized $C'$ enforces the new goals.                                   |
| GR(1) updating | Old/new GR(1) specifications and switching condition                                                           | New controller $C_2$ , bridge and switching bookkeeping                                                  | Selected $C_2$ runs unchanged after the bridge.                            |
| FG-DUCS        | Fixed components/controllers; local transfers; monitors, initializers and precedence; loadable targets         | Mixed-version and boundary game; policy, rank and handover map for every old entry and permitted outcome | Supplied new closed loop is preserved from the selected compatible target. |

*Independent requirement boundaries.* DUCS’s transition requirement  $T$  already expresses overlap, gaps and gap duties for one global stop/start pair [33, Sec. 2, Def. 4.1]. FG-DUCS separates each old stop and new start/initializer, as Policy(2)’s audit overlap and role exclusion require.

*Cell through native interfaces.* DUCS and GR(1) express Cell’s paths with added structure. Technical Appendices H–I give the supplied stages, lifetime predicates and complete constructions; these are choices for Cell, not mandatory native input formats. The DUCS witness retains every endpoint edge after its added handover  $h$  and requires eventual  $h$ , beyond the published safety- $T$  implementation [33, Sec. 5.3]; GR(1) may select fewer endpoint edges when synthesizing  $C_2$ . These controllable, deterministic examples imply neither general translation nor cost advantage. S5 compares the native contracts.

## 3 Defining an Update Contract

### 3.1 Supplied Choices

*Finite LTSs and safety monitors.* A labeled transition system (LTS)  $\mathcal{L} = (S, \Sigma_{\mathcal{L}}, \delta, s^0)$  has nondeterministic transitions  $\delta : S \times \Sigma_{\mathcal{L}} \rightarrow 2^S$ . Partition events as  $\Sigma = \Sigma_{uc} \uplus \Sigma_c$ : controllers may disable

controllables but cannot select outcomes or suppress uncontrollable (UC) events. A synchronous-product event requires a transition in every component whose alphabet contains it; other components retain their state [24]. A monitor is an automaton that checks a requirement as events occur. Requirement  $r$  has a total deterministic monitor  $t_r = (M_r, \Sigma, \eta_r, m_r^0, F_r)$  with absorbing errors for violating prefixes [1, 21]. The empty monitor product has one non-error state. Monitors observe labels; policies also observe successor component states.

*Components and transfers.* For each  $i \in I$ , let  $e_i^{\text{old}}, e_i^{\text{new}}$  be the pre-update and post-update components and let  $g_i \subseteq S_{e_i^{\text{old}}} \times S_{e_i^{\text{new}}}$  be their state-transfer relation. Define  $g_i(s) = \{s' \mid (s, s') \in g_i\}$ . The supplied  $g_i$  includes every modeled transformation outcome. Let  $E^v = \parallel_{i \in I} e_i^v$ .

*Requirements and activation.* Let pairwise-disjoint  $R^{\text{old}}, R^{\text{new}}, R^{\text{upd}}$  contain the pre-update, post-update, and update-interval safety requirements, respectively. For each  $r \in R^{\text{new}} \cup R^{\text{upd}}$ , the input supplies a partial *monitor-initialization map*  $\iota_r$  from version-tagged component tuples to  $M_r \setminus F_r$ . Its domain specifies where activation is allowed. Section 4.1 defines history-based and activation-scoped interpretations. For  $r \in R^{\text{upd}}$ ,  $\text{dom } \iota_r$  includes every reachable old component state.

*Endpoints and handover targets.* For  $v \in \{\text{old}, \text{new}\}$ , the supplied finite controller  $C^v$  preserves uncontrollables and makes  $E^v \parallel C^v$  deadlock-free and satisfy  $R^v$ . Hiding controller auxiliary events leaves observation alphabet  $\bigcup_i \Sigma_{e_i^v}$ ; the union over both versions is the ordinary alphabet  $\Sigma_N$ . Write  $\text{Reach}$  for initial-state reachability and

$$\text{CL}^v = E^v \parallel C^v \parallel \parallel_{r \in R^v} \iota_r.$$

Monitors are passive: every endpoint transition originates in  $E^v \parallel C^v$ ; a monitor only updates on that label and adds no enabled event. The input supplies nonempty, checked targets  $Z_{\text{load}} \subseteq \text{Reach}(\text{CL}^{\text{new}})$  with component, controller and new-monitor states.

*Update commands and order.* The set of update events, each executed exactly once, is

$$\Sigma_{\text{upd}} = \{\rho_i \mid i \in I\} \cup \{\text{stop}(r) \mid r \in R^{\text{old}}\} \cup \{\text{start}(r) \mid r \in R^{\text{new}}\}. \quad (1)$$

These controllable events transfer components, stop old monitors or start new ones, respectively, and are disjoint from ordinary events  $\Sigma_N$ :  $\Sigma = \Sigma_N \uplus \Sigma_{\text{upd}}$  and  $\Sigma_{\text{upd}} \subseteq \Sigma_c$ . Each transfer event  $\rho_i$  is one atomic model transition: the controller chooses the update, but cannot choose its  $g_i$  outcome.

An optional strict partial order  $< \subseteq \Sigma_{\text{upd}} \times \Sigma_{\text{upd}}$  expresses precedence among update operations. For a pending-event set  $P$ , event  $a$  is order-enabled exactly when  $a \in P$  and every  $a' < a$  has left  $P$ . Monitors self-loop on update events that they do not specify.

*Eligible contracts.* An eligible contract satisfies the finiteness, typing, totality and non-error conditions above; dependencies form a strict partial order, and endpoint controller alphabets cover environment labels. Eligibility is structural; Section 4.1 justifies initializer meanings.

### 3.2 States, Transitions, and Completion

*States and all entry states.* An update state  $q = (\xi, \mu, P)$  records each component's version/state  $\xi(i) = (v, s)$  with  $s \in S_{e_i^v}$ , each monitor's state  $\mu(t_r) \in M_r \cup \{\perp\}$  ( $\perp$  means inactive), and pending commands  $P \subseteq \Sigma_{\text{upd}}$ . Here  $\rho_i \in P$  exactly when component  $i$  is old. Old monitors remain active until their stop leaves  $P$ ; new monitors become active when their start leaves  $P$ ; update monitors are always active.

Admission  $\text{Init}(z)$  copies  $z$ 's component and old-monitor states, installs interval initializers, leaves new monitors inactive and sets  $P = \Sigma_{\text{upd}}$ . The update policy replaces  $C^{\text{old}}$  and its scheduling

restrictions. Declared old monitors remain until their stops; undeclared objectives, including old liveness, are not preserved. Duties needed throughout belong in interval monitors. The entry set is

$$Q_0 = \{\text{Init}(z) \mid z \in \text{Reach}(\text{CL}^{\text{old}})\}. \quad (2)$$

All-entry coverage forbids waiting for a favorable old state.

*Four kinds of induced transitions.* For  $\xi(i) = (v, s)$ , component  $i$  uses  $e_i^v$ . The active-component product determines enabled ordinary uncontrollables  $U(q)$ , including those with unsafe outcomes, independently of updates and goals. For every  $a \in \Sigma_{\text{upd}}$ , impose

$$U(q) \neq \emptyset \implies \text{Post}(q, a) = \emptyset. \quad (3)$$

This global *uncontrollable quiescence* differs from transaction quiescence [20] and tranquility [41]. Ordinary controls may coexist with UC. The following rules define  $\text{Post}$ ; unmentioned coordinates stay unchanged. “Advance” means  $\mu'(t_r) = \eta_r(\mu(t_r), a)$  for the specified active monitors.

**Ordinary event**  $a \in \Sigma_N$ . If  $a$  belongs to the local event set of at least one active component and every active component having  $a$  can transition, take every synchronous-product outcome and advance every active monitor.

**Component update**  $\rho_i$ . If  $U(q) = \emptyset$ , order-enabled, and  $\xi(i) = (\text{old}, s)$ , set  $\xi'(i) = (\text{new}, s')$  for each  $s' \in g_i(s)$ , delete  $\rho_i$  from  $P$ , and advance every active monitor.

**Requirement deactivation**  $\text{stop}(r)$ . If  $U(q) = \emptyset$  and order-enabled, advance every active monitor except  $t_r$ , retain inactive monitors at  $\perp$ , set  $\mu'(t_r) = \perp$ , and delete  $\text{stop}(r)$  from  $P$ ;  $t_r$  does not observe it.

**Requirement activation**  $\text{start}(r)$ . If  $U(q) = \emptyset$ , order-enabled, and  $\xi \in \text{dom } t_r$ , advance every active monitor, retain other inactive monitors at  $\perp$ , install  $\mu'(t_r) = t_r(\xi)$ , and delete  $\text{start}(r)$  from  $P$ ;  $t_r$  begins online observation afterward.

If a precondition is false, the successor set is empty; every nondeterministic outcome is included and environment-chosen. Let  $\mathcal{Q}$  contain every state reachable from  $Q_0$  through these transitions. *Safety and handover-ready goals.* An update state  $q = (\xi, \mu, P)$  is safe, written  $\text{Safe}(q)$ , exactly when every active monitor is outside its error states. For  $P = \emptyset$  and  $z \in Z_{\text{load}}$ , write  $q \sim z$  when component states (without version tags) and all new-monitor states agree. The handover map  $\kappa$  selects such a full endpoint state  $z$ . Goals are

$$Q_{\text{goal}} = \{q = (\xi, \mu, \emptyset) \in \mathcal{Q} \mid \text{Safe}(q) \wedge U(q) = \emptyset \wedge \exists z \in Z_{\text{load}} : q \sim z\}. \quad (4)$$

At a goal, interval obligations end and matched new monitors continue. Handover disables ordinary controllables and installs  $\kappa(q)$ ’s controller state, outside  $\Sigma$  and the rank. UC cannot compete; compatible targets may have different continuations.

*State-dependent policies and finite completion.* A memoryless update policy is a function  $\sigma : \mathcal{Q} \setminus Q_{\text{goal}} \rightarrow 2^{\Sigma_c}$  selecting only events with nonempty successor sets.

The possible next states under this policy are

$$\text{Succ}_{\sigma}(q) = \bigcup_{u \in U(q)} \text{Post}(q, u) \cup \bigcup_{a \in \sigma(q)} \text{Post}(q, a). \quad (5)$$

The environment chooses any uncontrollable and any permitted-event outcome. The returned policy retains one controllable at a quiescent state.

Let  $\text{Reach}_{\sigma}(Q_0)$  be the policy closure from  $Q_0$ , stopping at goals.

*Runs, winning, and witnesses.* A run follows the policy successors and stops at a goal. It is *maximal* if infinite or ending at a goal or a state without successors. A single policy  $\sigma$  is winning for an FG-DUCS instance if, from every  $q_0 \in Q_0$ , (i) every reachable update state is safe, (ii) every reachable non-goal state has a successor, and (iii) every maximal update run reaches  $Q_{\text{goal}}$  after

finitely many events. This *finite completion* counts events and assumes no fairness. Winning policies exclude controllable self-loops during update; a reached UC self-loop makes a state losing. Ordinary actions can progress, as Cell’s moves and inspection do. Handover restores the supplied controller’s full matched behavior.

A *progress-ranking function* for a winning policy is  $\text{rk} : \text{Reach}_\sigma(Q_0) \rightarrow \mathbb{N}$ , with value 0 at goal states and satisfying  $\text{rk}(q') < \text{rk}(q)$  for every non-goal state  $q$  and every  $q' \in \text{Succ}_\sigma(q)$ .

*The FG-DUCS problem.* Decide whether one policy safely completes the eligible game from every entry under every permitted event and outcome. Return  $(\sigma, \text{rk}, \kappa)$  covering all  $Q_0$ , or a losing-region certificate.

### 3.3 Why the Generated State Represents the Contract

A *local update history*  $\ell = z(a_1, o_1) \cdots (a_n, o_n)$  records the old entry, events and realized local successors. Components follow their active-version transitions and transfers; monitors fold their declared active intervals. Commands obey precedence, guards and UC quiescence. A source policy selects controllable labels, retains every UC and permitted outcome, and safely completes every maximal history at a matching quiescent target. Technical Appendix A defines these contract operations on histories.

**PROPOSITION 3.1 (ADEQUACY OF THE GENERATED UPDATE GAME).** *For an eligible contract, map each local update history to its current component tuple, active monitor memories and unexecuted commands. Histories with the same image admit the same next events and all the same local outcomes, agree on safety, and have the same handover targets. This map gives a label- and outcome-preserving correspondence between local update histories and paths of the generated game. A history-dependent policy satisfying the source obligations exists exactly when a winning policy exists in that game.*

**PROOF.** Fold a source history into its current tuple  $F(h)$ . At entry this is  $\text{Init}(z)$ . Each ordinary step uses the same participating component edges and full Cartesian product of outcomes; each transfer uses the same quiescence/order tests and  $g_i$ , advances active monitors and consumes its command. Stop removes its own monitor before observation; start installs its initializer after observation by existing monitors. Thus every label and outcome has exactly the same successor image, and safety and matching targets agree.

Induction maps source histories to game paths. Conversely, choose an old tuple witnessing the entry and lift each edge by this one-step correspondence, also for infinite paths. A game policy lifts through  $F$ ; applying a source policy to consistently lifted game histories gives a history-dependent game policy. Both directions preserve UC events, all selected-label outcomes, unsafe prefixes, deadlocks and infinite avoiding runs. Finite reachability attractor layers yield a state-based winner exactly when any history-dependent winner exists (the independent memoryless-policy lemma in Technical Appendix A), completing the equivalence.  $\square$

This proves correspondence for the supplied contract; requirement meaning and Java frontend correctness remain separate obligations.

**PROPOSITION 3.2 (PRESERVATION OF THE SELECTED ENDPOINT).** *For a returned WIN certificate and reached goal  $q$ , the new-phase submodel reachable after handover at  $q$  is label-preserving isomorphic to  $\text{CL}^{\text{new}}$  rooted at  $\kappa(q)$ . Hence their finite and infinite ordinary-event traces coincide.*

**PROOF.** Handover retains matching components and monitors and installs the selected controller state. Each new-phase state copies an endpoint state with exactly its labeled transitions, without further policy restriction or return to updating. This bijection preserves and reflects edges, hence finite and infinite paths.  $\square$

Continuation properties established at the selected state are preserved; target selection can change which continuation applies.

*Why matching and reconnection both matter.* Cell’s prefix  $\rho_B, m_B, \text{start}, \text{stop}, \rho_A$  finishes every command with inspection pending. Physical matching alone admits clear target  $z_B$ , whose  $m_A$  violates that duty; monitor matching excludes the reset. Conversely, disabling controllable loop  $b$  in a one-state endpoint with loops  $a, b$  and goal *true* preserves legality, deadlock freedom and safety [33, Defs. 3.9–3.10, 4.3], but loses  $b$ -continuations. Reconnection preserves them.

Technical Appendix G characterizes policy reuse under changed endpoints, with a saved PC2 specialization.

#### 4 What a Winning Policy Guarantees

Initial monitor states must represent current invariants or history-dependent duties. Table 3 separates requirement meaning, game guarantees and execution.

Table 3. Guarantee scopes and their premises. RQ2 reports validation coverage.

| Scope                | What follows                                                        | Additional premise or boundary                                                  |
|----------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Game                 | Safe finite completion for all entries/outcomes; matched handover   | The supplied transition and endpoint rules (Theorem 6.1)                        |
| Invariant from start | Truth at activation and while active                                | Correct predicates/observations and initializer conformance; ACT below          |
| Safe prior histories | Continuations preserve obligations of the admitted safe histories   | Residual inclusion; an empty history scope makes inclusion vacuous              |
| From update entry    | UPD monitors reflect obligations from admission onward              | Correct boundary state and entry interpretation; S3 gives sufficient conditions |
| Execution            | Handover within the entry’s event rank; selected endpoint continues | A1–A4; no elapsed-time bound or guarantee of platform conformance               |

Source checks alone do not establish actual activation conformance. Cell’s occupancy duty is proved below; S3 treats stronger whole-history meaning separately.

##### 4.1 Giving Requirements Their Intended Lifetimes

An initializer assigns the obligation present at activation; selecting any non-error state is insufficient.

*An exact activation obligation.* Cell’s owed-inspection bit starts at 1 exactly when  $B$  holds the workpiece. Event  $m_B$  sets it,  $j$  clears it, and  $m_A$  is forbidden while it is 1; other labels preserve it. Encoding 0 by  $c$  and 1 by  $d$  makes every monitor transition identical to this rule, including rejection of forbidden departure. Initialization and suffix induction prove exact enforcement. Figure 1’s first path starts with debt and clears it by ordinary inspection; the second starts clear.

*Invariants that begin at activation.* Activation-scoped interpretation ACT checks a pure invariant  $\Box\varphi$  at the actual post-start observer vector and after every later label. It retains factual fluent values, not a past-violation bit. For deterministic observers with total update  $d : V \times \Sigma \rightarrow V$ , define

$$\text{ACT}_\varphi(v) = \{w \mid \forall u \leq w : \varphi(d^*(v, u))\},$$

where prefixes include the empty word. A faithful safety monitor  $M$  has non-error language  $S_\varphi = \text{ACT}_\varphi(v_0)$  from its initial state. Let  $L_M(m)$  denote its continuations from state  $m$ .

PROPOSITION 4.1 (SAFE LOOKUP CELLS FOR PURE INVARIANTS). *If a lookup at  $v$  selects a non-error state  $m$  reached together with  $v$  by a safe word in the unrestricted monitor–observer product, or a state with the same continuation language, then  $\varphi(v)$  holds and  $L_M(m) = \text{ACT}_\varphi(v)$ .*

PROOF. Let  $h$  reach the pair  $(m_h, v)$ . Faithfulness and non-error reachability give  $h \in S_\varphi$ , hence  $\varphi(v)$ . For every suffix  $w$ ,

$$w \in L_M(m_h) \iff hw \in S_\varphi \iff w \in \text{ACT}_\varphi(v).$$

The second equivalence uses safety of the prefix  $h$  and deterministic observers: future valuations depend only on  $v$  and  $w$ . Language equivalence transfers the equality to  $m$ .  $\square$

The witness need not be plant-executable or obey one-shot update restrictions. This is residual-language reasoning [12, 36], applied to the generated lookup. At activation, the adapter must choose the actual post-start key and use the same future observation rules; start is consumed once. Undefined entries and fallback initialization are outside the criterion. A true  $\varphi(v)$  alone does not supply a lookup entry.

RQ2 distinguishes finite source checks from conformance at actual activations (Technical Appendix E.1).

*When prior histories are part of the requirement.* For prefix-closed requirement language  $S_r$ , declared activation histories  $H_r(\xi)$  and monitor continuations  $L_r(m)$ , residual soundness (RS) means  $L_r(l_r(\xi)) \subseteq \bigcap_{h \in H_r(\xi)} \{w : hw \in S_r\}$ . New histories include start once; monitors consume later events. Equality is exact; strict inclusion can reject feasible updates, and an empty history set makes inclusion vacuous. Scope completeness and intent are external obligations (Technical Appendix B.1).

The saved checks use safe pre-activation histories consistent with the current observation for new requirements and a reference beginning at entry for interval requirements. A whole-history interpretation including prior violations is not claimed (S3). Cell’s pending initializer is exact for occupancy-based debt but conservative for a start-only reference that starts clear. Safe-history inclusion alone establishes neither the occupancy duty nor ACT.

## 4.2 Execution and Handover

The execution connection has four distinct premises (Technical Appendix F):

**A1:** endpoint observations preserve component/monitor states and same-labeled endpoint edges.

**A2:** update control disables only unselected controllables, preserves every UC event/outcome, reports the successor and eventually observes a next event.

**A3:** commands start quiescently and finish atomically, finitely and failure-free with an observed modeled successor.

**A4:** entry observes/copies the actual old state; quiescent handover installs the selected compatible state, using finite failure-free linearizable boundaries [16].

Under A1–A4, every execution maps to the linked model and reaches handover within the entry rank (Theorem F.1). History completeness places every actual activation history in its declared scope; with RS it justifies the residual interpretation. Internal divergence and unobservable outcomes are excluded; no fairness is assumed and rank bounds events, not time.

## 5 How Granularity Changes Feasibility

Merging commands removes intermediate states. We show why this can either prevent or enable completion; Section 7.2 evaluates the specified comparisons.

What is merged. Partition transfers, old stops and new starts separately into blocks. A transfer block  $B \subseteq I$  simultaneously permits all targets in  $\prod_{i \in B} g_i(s_i)$ ; a boundary block removes or initializes its monitors at the same component tuple. Other active monitors compose member-label effects, checked to commute pairwise on every monitor/residual state. Mapped precedence must remain irreflexive and acyclic. Ordinary behavior, entries, endpoints, relations, initializers and targets stay fixed; quiescence and all member guards apply. These are comparison contracts, not DUCS encodings.

LEMMA 5.1 (MANDATORY-COMMAND OBSTRUCTION). *Let  $q_0$  be an entry and  $B$  a nonempty set of commands pending there. Suppose a safe region  $R$  contains  $q_0$ , retains all commands in  $B$ , and is closed under every safe successor whose label is outside  $B$ . If each command in  $B$  is disabled or has an unsafe outcome at every state in  $R$ , no winning policy covers  $q_0$ .*

PROOF. Every goal exhausts pending commands, so a winner must execute a first command in  $B$ . Safety and closure keep its prefix in  $R$ , where that step is disabled or has an environment-chosen unsafe outcome. Avoiding it yields a non-goal deadlock or infinite run, also losing.  $\square$

In Cell’s one-workpiece region, joint transfer is disabled. Policy retains its initial active bits until either grouped boundary violates the interval. PC2-Rolling’s enabled joint transfer violates availability (Section 7.2.1). These results concern specified merges, not native DUCS expressiveness.

A capacity boundary. Readiness gates motivate the constructed Rolling contract [31, 38]. It starts with  $n$  ready replicas, no spares, and an interval floor of  $m$  ready replicas. A transfer boots its replicas; each becomes ready after one UC event, with no waiting loop. Service self-loops are controllable. The all-new, ready target restores capacity; service may pause during the update.

PROPOSITION 5.2 (ROLLING THRESHOLD). *For  $n \geq 2$ ,  $1 \leq m < n$  and  $1 \leq k \leq n$ , fixed consecutive batches of size  $k$  (with a smaller final remainder) admit safe finite completion iff  $k \leq n - m$ .*

PROOF. Some mandatory batch has size  $k$ . Its transfer leaves at most  $n - k$  ready replicas, so  $k > n - m$  violates the interval. Otherwise, transfer one batch only when all replicas are ready, disable service, and retain every UC readiness outcome before the next transfer. Readiness never falls below  $n - k \geq m$ . With  $u$  old and  $b$  booting replicas,  $2u + b$  strictly decreases on every selected transfer and every UC event, reaching zero exactly at the target. Thus all maximal runs finish without fairness.  $\square$

Why finer is not always better. Two components with one state per version share UC labels  $x, y$ . The only UC loops are  $y$  in old 1,  $x$  in new 1,  $x$  in old 2 and  $y$  in new 2; shared labels synchronize. Thus all-old and all-new are quiescent, but either mixed product has a UC loop. With identity transfers and no requirements, either first local transfer prevents quiescence forever, so fine loses. A joint transfer skips the mixed state and wins at rank one. A shared controllable loop makes both endpoints deadlock-free. Atomicity can therefore remove either a necessary or a harmful intermediate state; there is no general refinement order (Technical Appendix O.12).

## 6 Synthesizing and Checking an Update Policy

The procedure specializes strong-planning and on-the-fly reachability principles [7, 8, 40], retaining every queried event outcome and producing evidence for every old entry.

A bucket is  $\text{Post}(q, a)$ ; empty means disabled. Disabling controls at a UC-enabled state removes runs without creating deadlock, preserving winning (UC priority, Lemma A.1). With quiescent goals and no fairness or controller priority, permitting commands while UC is enabled therefore adds no winning entries. Changing fairness or priority changes the game. The proof establishes memoryless sufficiency.

## 6.1 Lazy Successor Generation

Algorithm 1 constructs a reachability attractor [5].  $V$  contains discovered states,  $X$  those with all UC candidates queried, and  $W$  proved winners. A UC state needs all successors to win; a quiescent state needs one complete control bucket. Lazy queries controls incrementally, keeping unfinished cursors in Retry. Each promotion records a complete nonempty proving set and a rank above its members. Reverse incidences propagate promotions. Failure requires exhausted candidates and a promotion fixed point. Extraction chooses lower-rank controls and compatible targets; it need not minimize path length.

---

### Algorithm 1: Lazy

---

**Input:** Typed update contract (Section 3)

**Output:**  $(\sigma, \text{rk}, \kappa)$  or (no solution,  $L$ )

```

1  $V \leftarrow Q_0$ ;  $X, \text{Retry} \leftarrow \emptyset$ ;  $W \leftarrow V \cap Q_{\text{goal}}$ ;
2  $\text{Open} \leftarrow \{q \in Q_0 \mid \text{Safe}(q) \wedge q \notin Q_{\text{goal}}\}$ ; set  $\text{rk}(q) \leftarrow 0$  for  $q \in W$ ;
3 while true do
4   Add queued goals to  $W$  once with rank zero; exhaust Promote, retaining witnesses/ranks, removing winners
   from Retry and propagating predecessors;
5   if  $Q_0 \subseteq W$  then return  $(\sigma, \text{rk}_{|\text{Reach}_\sigma(Q_0)}, \kappa)$  constructed from the goal witnesses and retained lower-rank
   choices;
6   if  $\text{Open} \neq \emptyset$  then
7     Remove  $q$  from Open, add it to  $X$ , and query all uncontrollable candidates;
8     Close the control cursor if UC-enabled; otherwise initialize it and query to the next nonempty bucket or
     exhaustion;
9     Insert  $q$  into Retry iff  $q \notin W$  and its cursor has remaining candidates;
10  else
11    if  $\text{Retry} = \emptyset$  then return (no solution,  $L \leftarrow V \setminus W$ );
12    Remove  $q$  from Retry; query to the next nonempty bucket or exhaustion; reinsert iff  $q \notin W$  and its
    cursor has remaining candidates;
13  Add every queried successor to  $V$ ; queue newly discovered goals for rank zero and safe non-goals for Open
    (unsafe states stay outside  $W$  and Open);
```

---

## 6.2 Correctness and Certificates

**THEOREM 6.1 (TERMINATION, SOUNDNESS, AND COMPLETENESS).** *For every eligible finite FG-DUCS instance with Equation (3) and quiescent goals in Equation (4), Algorithm 1 terminates. A returned  $(\sigma, \text{rk}, \kappa)$  keeps every active monitor non-error, preserves every uncontrollable event and every outcome of each policy event, reaches  $Q_{\text{goal}}$  finitely without deadlock, and gives each reached goal a compatible target in  $Z_{\text{load}}$  for the supplied game. It returns no solution iff no winning policy covers  $Q_0$  in that supplied game.*

**PROOF.** *Coverage.* Every new safe non-goal enters *Open* and leaves only after expansion queries all UC candidates. At quiescence, finite control cursors retain unfinished nonwinners in *Retry*; promotion removes only winners. Thus expanded states have complete UC outcomes, unfinished nonwinners remain retryable, and unexpanded safe non-goals remain open.

*Termination.* Each state is expanded/promoted once and each candidate queried once. Reverse incidences are processed on promotion, so the finite search terminates.

*Soundness.* Goals start at rank zero with compatible handover states. By induction, each promotion adds a safe state above a complete nonempty proving set of previously ranked winners: every UC successor, or every outcome of one control event. Queried buckets are immutable. Extraction

retains all UC events, chooses one such control bucket at quiescence, and disables other controls. Every reached non-goal is safe, has a successor, and every successor has strictly smaller natural-number rank. Consequently no maximal run is an infinite run or a non-goal deadlock, and every run reaches a matching quiescent goal. The monitor rules advance exactly the memories active over each event; induction from entry establishes every declared safety interval. Handover retains the matching new memories.

*Completeness.* At failure both exploration queues are empty and promotion has reached a fixed point. Some entry is in  $L = V \setminus W$ , and  $L$  contains no goal. Coverage ensures that each safe state in  $L$  was expanded. If it is UC-enabled, at least one UC outcome lies in  $L$ , since otherwise it would be promoted. At quiescence every control candidate is exhausted; each enabled control bucket has an outcome in  $L$  for the same reason. Against any history-dependent policy the environment can select such an outcome after every history. Disabling all events deadlocks; otherwise the resulting run reaches an unsafe state or stays outside the goals forever. Hence no winning policy covers that entry. A winning policy therefore rules out failure and, by termination, forces success. Finally, UC priority loses no winning policy: removing optional controls at a UC-enabled state removes runs without introducing a deadlock, and the restricted policy remains admissible in the original game.  $\square$

*Checking the returned evidence.* WIN checking requires exactly  $D_\sigma = \text{Reach}_\sigma(Q_0)$  as the rank domain, safety, enabled selected controls, nonempty all-outcome closure and strict decrease at non-goals. Goals require rank zero, quiescence and a compatible  $\kappa$ , defined exactly on  $D_\sigma \cap Q_{\text{goal}}$ ; omitted policy entries are empty. LOSS checking requires an entry-containing, goal-free  $L$  with the UC/all-control counter-outcomes used in the completeness proof. These certify search relative to the successor and endpoint predicates [18]; Section 7 states their implementation trust boundary.

### 6.3 Exploring the Compiled Contract on Demand

With indexed predicates and constant-time bookkeeping, search is linear in discovered states, queried candidates and returned outcomes, excluding preprocessing and Java sorting/scans. Lazy may explore the whole reachable graph. For independent one-state updates, transfer-first Lazy instead visits  $n + 1$  states versus Direct-Full's  $2^n$  subsets; Technical Appendices K and O give the accounting and complete default-order proof.

## 7 Implementation and Evaluation

### 7.1 Design and Implementation

RQ1 tests granularity mechanisms; RQ2 checks games, certificates and source monitors; RQ3 compares exploration and cost on 27 adapted contracts. Auxiliary trials remain outside fixed-budget medians.

The 9 inherited inputs each have three variants. Base has no interval requirement; R1 and R2 are alternatives. GSM/R1 blocks communication and decoding while a flag is set by old-stop and cleared by new-start. GSM/R2 requires both decode-rule boundaries before the environment transfer, without ordering the boundaries. Restricted actions and boundary/transfer pairs vary by model. PC2-Rolling is a separate source-derived availability case. Legacy also synthesizes subsequent control, a different objective.

*Implementation.* Our Modal Transition System Analyser (MTSA) extension [10] checks certificates [28]; Link checks old-update-new composition, entries and targets. An independent oracle parses restricted Finite State Processes (FSP). Finite-JSON runs save losing membership; FSP benchmarks report only its statistics.

*Use of generative AI.* Generative AI assisted contract and example design, proof drafting, Java solver/frontend implementation, Python checkers, experiment drivers, execution, result analysis and manuscript editing. Validation used certificate and exhaustive checks for algorithms, source and automaton comparisons for initialization, saved-output checks for tables and primary sources for citations. Claude Opus 5.5 and ChatGPT Pro/Latest supplied editorial reviews, not empirical evidence. Authors retain responsibility; shared AI/code can produce correlated faults. Technical Appendix P details tool roles; no human evaluation was performed.

*Constructing the benchmark inputs.* Nine inputs adapt eight DUCS sources; PC1/PC2 share a parent. PC1/PC2 preserve its repeatable, state-gated arrivals while making tool results UC. With work/output withheld, pending tool results and reset/arrival chains finish in quiescent arm states. Other adaptations shorten Railcab transfers preserving both outcomes, omit PowerPlant’s initial ENV pair, and change safety/ordering. Technical Appendix O details these changes; equivalence is unestablished.

From FSP declarations, MTSA synthesizes fixed endpoints and admits every reachable new-endpoint state. Before search, it compiles monitors and global-tuple initializer tables over physical states with observer values; UPD uses a constant post-entry state. Observers track nonlocal labels and survive commands/transfers. Finite-JSON Cell<sub>n</sub> inputs instead supply holder-dependent initializer maps directly (S3). Lazy defers update-game exploration; S4 retains preparation times and tuple counts for all 27 contracts.

Inputs have 1–5 components, 4–131 entries and 3–26 commands (Technical Appendix L). Only Railcab transfers branch; none declares precedence. R2 encodes order in monitors. GSM/PC1’s single transfer makes merging the identity.

*Exploration variants.* All methods share entries, successors, goals, checks and terminals. On-the-fly (OTF) variants omit controls at UC-enabled states and stop when all entries win. Direct-Full constructs all reachable buckets, including those controls, then applies UC priority; auxiliary Full+UC omits them during construction. At quiescence, Lazy queries individual buckets, Eager all controls; Update-first tests O-DUCS’s update priority [17] within our solver. While stops remain, Lazy tries stop, transfer, then ordinary buckets. It next favors remaining transfers; only when both are exhausted does an endpoint match put start first. This orders queries, not required executions; LIFO expansion precedes FIFO retries (full state/tie rules in Technical Appendix K).

*Design and measures.* RQ3 uses one Windows/Xeon host, 64 GiB heap and 1,200 s JVM cap: 27×4 same-game cells plus 27 Legacy references. Medians require five completions; censored cells retain their capped trial. Solver time includes checking; total JVM adds frontend, startup, output and endpoints. We also report exploration, policy size and peak working set. Legacy uses its native timer; Mac pilots are excluded. Full settings and interruptions remain in Technical Appendices P and O.8 and S4.

## 7.2 RQ1: Can Local Choices Change Feasibility?

We compare inherited inputs, constructed families and a source-derived case under Section 5’s merging rules. These comparisons do not estimate deployed prevalence.

*Inherited inputs: no observed change.* On nine inherited inputs, merging transfers, boundaries or both changes 0/54 Base/R1 decisions. Their total raw transfers omit Rolling’s startup phase; totality alone does not ensure safe merging. States decrease in 44 comparisons and stay equal in 10. R2 names individual events and is excluded. This null result limits observed granularity benefits.

*Constructed mechanisms.* The families isolate local choices, retaining both-WIN and both-LOSS controls.

Table 4 gives each mechanism and all outcomes. The 15 Rolling separations compare  $k = 1$  with  $k = n$ ; the 70 threshold cells are separate. Of the 33 separations, 30 are Rolling/Canary parameter points, not independent mechanisms. Threads tests both-WIN/both-LOSS quiescence controls. Technical Appendix O.11 gives all definitions and assumption controls.

**Answer.** Capacity loss, skipped intermediate stages and incompatible boundary orders yield 33/55 fine-WIN/merged-LOSS comparisons across five constructed families. Inherited decisions do not change. PC2’s source proof establishes merged impossibility, but its merged timeout leaves the solver comparison incomplete.

Table 4. Granularity mechanisms, parameter sweeps and controls. Each pair compares fine and merged contracts of one model: Policy merges boundaries, other rows merge transfers. Counts are comparisons, not independent applications. The main total excludes Cell.

| Family / control                                 | Mechanism and parameters                                           | Pairs | WIN /<br>LOSS | LOSS /<br>LOSS | WIN /<br>WIN | Inc. |
|--------------------------------------------------|--------------------------------------------------------------------|-------|---------------|----------------|--------------|------|
| <i>Constructed families and parameter sweeps</i> |                                                                    |       |               |                |              |      |
| Rolling                                          | Readiness during startup.<br>$n = 2 \dots 6, 1 \leq m < n$         | 15    | 15            | 0              | 0            | 0    |
| Canary                                           | Failure reports and recovery.<br>$n = 2 \dots 6, 1 \leq m < n$     | 15    | 15            | 0              | 0            | 0    |
| Policy                                           | Audit overlap; role exclusion.<br>2 rule pairs                     | 1     | 1             | 0              | 0            | 0    |
| DB-Rolling                                       | Role-gated replacement.<br>$n = 3, m = 2, 3$                       | 2     | 1             | 1              | 0            | 0    |
| Rolling+Audit                                    | Readiness and audit overlap.<br>$n = 2, m = 1$ ; 1 audit pair      | 1     | 1             | 0              | 0            | 0    |
| <i>Quiescence control</i>                        |                                                                    |       |               |                |              |      |
| Threads (control)                                | Backpressure or persistent arrival.<br>$n = 2 \dots 6$ ; queue 1,2 | 20    | 0             | 10             | 10           | 0    |
| <i>Source-derived attempt</i>                    |                                                                    |       |               |                |              |      |
| PC2-Rolling                                      | Availability during calibration.<br>2 arms; floor 1                | 1     | 0             | 0              | 0            | 1    |
| Main total                                       |                                                                    | 55    | 33            | 11             | 10           | 1    |
| Cell <sub>n</sub> (reference)                    | Empty-station transfer guards.<br>$n = 2 \dots 10$                 | 9     | 9             | 0              | 0            | 0    |

Outcomes read fine / merged; Inc. is unresolved. PC2’s merged solver remains TO, not LOSS. Rolling and Canary share a capacity obstruction: their sweeps supply 30 of the 33 WIN/LOSS pairs.  $n$  counts replicas/workers/stations. Canary bounds reported failures by  $n - m$ ; otherwise  $m$  is an availability floor. Threads varies queue capacity 1,2 under both arrival regimes. Rolling+Audit’s boundary merge remains WIN and adds no pair here.

**7.2.1 A Source-Derived Calibration Case.** PC2-Rolling adds calibration while requiring another arm operational. Requests are controllable, completions UC, and transfers enter calibration while

preserving initializer observers. Erasing calibration recovers the original operational process, without whole-problem equivalence. Endpoints retain baseline requirements plus availability; the interval additionally treats transfers as calibration entry. The 25 boundaries and two transfers have no supplied order (Technical Appendix J).

The saved policy calibrates arm 1, waits for completion, then calibrates arm 2 and waits again. All 126 entries are covered, every outcome decreases rank from at most 37, and at least 1 arm stays outside calibration. Link checks the full composition. Each of the thirteen source invariants holds at its start and through the remaining start-only suffix until handover, proving ACT on reached tuples. This covers neither all initializer tuples nor all 138 occurrences; GSM below exercises ordinary actions under an active new requirement.

History-wide obligations give a different result: only 39/126 entries admit a safe full prefix; 87 fail (21 before entry, 66 during update), and all admit an earlier violating cycle restoring zero observers. Safe-history nonemptiness for 38,288 initializer tuples and runtime remain unchecked. Technical Appendix J retains each source formula, activation path and history partition.

*Why the fixed merge loses.* Every old entry admits both raw transfers. Before joint transfer, old plants lack calibration actions and all other steps preserve readiness. The joint transfer therefore puts both arms in calibration, clearing both interval readiness bits and reaching error. This proves impossibility, while the merged solver and fine Direct-Full both time out at 7200 s/200 GiB. Those timeouts remain unresolved solver results.

*Additional witnesses.* Restricted-transfer comparisons retain both-LOSS and incomplete cases separately (Technical Appendix O; S1–S4).

### 7.3 RQ2: What Do the Checks Establish?

**Answer.** Table 5 maps the checks and their populations to the guarantee layers in Table 3.

Table 5. What each RQ2 validation population establishes.

| Check                | Population                                               | Established result and boundary                                                                                       |
|----------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Independent games    | 46 fixtures, two solvers: 92 jobs                        | 36 WIN, 48 LOSS, 8 invalid; all 84 eligible jobs pass certificate/exhaustive checks.                                  |
| NEW initialization   | 46 source positions, 138 occurrences; 900 retained cells | Exact start-scoped languages under reference observations (Proposition 4.1); actual activation conformance unchecked. |
| UPD initialization   | 135 occurrences: 46 R1, 89 R2                            | Exact entry-scoped initialization from source guards and stage fluents; execution is unchecked.                       |
| Application policies | 25 Lazy WIN contracts; 125 runs                          | Certificates check the shared Java game. Runs save summaries; complete PC2-Rolling/Railcab exports are separate.      |

*Independent reconstruction.* For the 92-job suite, Python reconstructs explicit successors; front-end and endpoint/goal code are shared. All WIN jobs pass Link; invalid cases omit events, deadlock or suppress UC. Six separate monitor-free fixtures independently reconstruct FSP endpoints, games and goals (S4.2).

*From source invariants to initialized monitors.* Initial truth, observer updates and monitor steps establish invariant fidelity by induction; safe lookup reachability supplies Proposition 4.1’s premise. These are static checks: actual decoding, post-start keys, subsequent observations and absence of fallback remain unvalidated. Technical Appendix E.1 retains every entry, ERROR case and one-step count; arbitrary FSP compilation is unvalidated.

*Which history is meant.* All 138 NEW occurrences also satisfy residual inclusion for safe prior histories; actual scope nonemptiness remains unverified except one PC2 tuple. UPD monitors instead begin their reference at update entry (Technical Appendix D).

*GSM after activation.* GSM changes  $\square\text{-decode.3}$  to  $\square\text{-decode.2}$ . A source-derived all-entry policy starts the new requirement with observer/monitor zero, then permits *decode.3* and output before transfer while rejecting *decode.2*. It reaches a matching endpoint within twelve events and satisfies R1’s gap restriction. This proves ACT over ordinary behavior; PC2’s active suffix contains only starts. The complete path, rank proof and separate 45-pair entry-initialization check are in Technical Appendix E; no new solver result is claimed.

*What application WIN and LOSS mean.* Railcab’s 22-entry Base policy permits 1–4 ordinary gap events; R1 has none and retains seven interval guards. Both pass certificate/Link. Duties needed between old-stop and new-start require interval requirements; WIN alone does not supply them (Technical Appendix O.9).

Industry/R1 loses at 42/131 entries: GF1/TOR validation violates new monitors; DSD1 permits endless rejection, or quiescent deadlock if its only safe validation is withheld. Exhaustive closure excludes every order. Native DUCS wins its different subsequent-control problem (Technical Appendix O; S4).

Lazy’s winning certificates have 8–1,071 update states and maximum ranks 4–34, consistent across five repeats. Linked controllers have 15–1,260 states (S4). Ranks bound events, not downtime or optimal length (Technical Appendix O.8).

#### 7.4 RQ3: Does Lazy Reduce Exploration?

**Answer.** Lazy avoids states and queries in all 14 completed Direct-Full pairs. Within the fixed budget it decides 26 contracts, Eager 17 and Direct-Full 14; all paired decisions agree (Table 6). Update-first is faster on 8/24 paired solver medians but adds two timeouts. Lazy returns LOSS for Industry/R1 and times out on PC2/R2 at 1,200 s/64 GiB (fixed), 3,600 s/64 GiB (sensitivity) and 7,200 s/200 GiB (S4, earlier series); the limiting resource is unknown.

Legacy yields 9 WIN, 13 OOM and 5 unmeasured snapshot-cap cases. Its differing subsequent-control objective precludes same-game ratios or impossibility inferences (Technical Appendix O).

Figure 2 plots all 14 completed pairs (full rows in Technical Appendix M). Base/R1/R2’s 5/3/6 pairs have median Direct-Full/Lazy state ratios 111.89/11.88/2.75. These different completed populations do not isolate interval requirements’ effects or attribute savings to old-stop/new-start gaps. In R2, Update-first uses fewer states and a lower solver median on six of eight pairs (Technical Appendix O).

Preparation remains substantial: Lazy’s adapter medians reach 14.221 s (PC2/R1; S4). Capped Direct-Full attempts give total-JVM lower bounds, not solver ratios or medians (Technical Appendix O.10).

*Separating construction choices.* Lazy versus Eager isolates deferred controllable-bucket generation under the same UC pruning, input, host and budget: 26 versus 17 decisions. Direct-Full additionally changes pruning. Auxiliary Full+UC timings cannot isolate that effect across differing builds, hosts and repetitions; no unpruned OTF is implemented (Technical Appendix O.10).

Table 6. All 27 contracts per method. TO denotes one capped trial, not a timing median. Ratios summarize heterogeneous completed pairs; values above one favor Lazy.

| Method       | WIN/LOSS/TO          | Faster / pairs | Solver ratio         | Total JVM ratio    |
|--------------|----------------------|----------------|----------------------|--------------------|
| Lazy         | 25/1/1               | —              | —                    | —                  |
| Eager        | 17/0/10              | 0 / 17         | 2.04 [1.18–4,575.05] | 1.60 [1.00–144.85] |
| Update-first | 23/1/3               | 8 / 24         | 1.07 [0.19–39.76]    | 1.00 [0.32–6.10]   |
| Direct-Full  | 14/0/13 <sup>†</sup> | 0 / 14         | 9.14 [1.58–2,123.81] | 2.27 [1.00–130.66] |

TO: timeout; no OOM/unmeasured cases. <sup>†</sup> Workflow/R2 retains its interruption and later same-setting timeout (S4). Comparator/Lazy ratios are median [min–max] across completed pairs of five-run medians. Faster / pairs counts lower comparator solver medians among all completed pairs. JVM includes frontend, startup, endpoints and output; solver includes checking. Unresolved costs have no ratio.

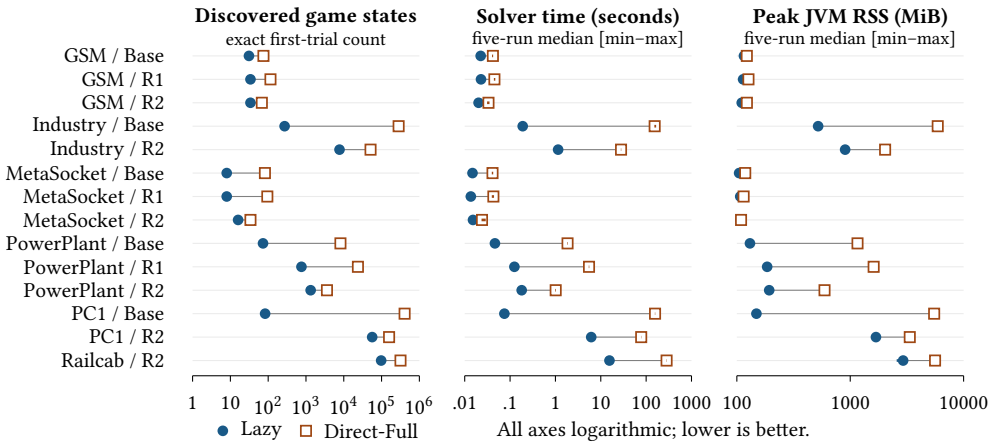


Fig. 2. All 14 completed WIN/WIN pairs. Solver time includes checking but excludes frontend, endpoint preparation and JVM startup. RSS is sampled resident memory over the whole JVM run, not solver-only or heap usage. The other 13 Direct-Full contracts time out (Figure A4) and are omitted. This compares whole procedures.

*Scaling limits.* All methods finish the independent family for  $n = 2, \dots, 20$ . At  $n = 20$ , Lazy and Update-first discover 21 states each, Eager 211, and Direct-Full 1,048,576 (S4).

Travel reaches timeouts and out-of-memory failures under all methods, including expanded-budget timeouts. These separate trials never enter medians; Technical Appendix O.10 retains all populations and adverse outcomes.

## 8 Related Work

*Generating update environments.* Yamauchi et al. generate an update environment from pre/post environments and supplied global state-preservation transitions [43, Sec. V]. Table 2 distinguishes FG-DUCS’s local input contract from existing update interfaces; this is not the first generated update environment.

*On-the-fly updating.* O-DUCS separates update and post-update analysis [17, Sec. 3]. S4.8 documents the archived fork’s plain OTF: it precomputes a new controller and reconnects matching environment/safety states [4]. Identity with the 2024 release is unestablished.

*On-demand synthesis.* DCS already uses on-demand component products, defers controllable alternatives, treats UC-enabled states as uncontrollable and propagates success/failure backward [8, Sec. IV]. Its 2016 formulation requires safety and eventual goal occurrence on every trace; OTF-DCS (2023) requires a marked extension of each finite controlled trace [9, Def. 4]. FG specializes these patterns to its generated contract, with complete outcome buckets and all-entry certificates. Compositional and modular approaches structure control specifications [30, 42].

*Supervisor switching.* State reinitialization enables safe, deadlock-free switching between supplied supervisors on one plant [29, Sec. 3.3]. Jiao et al. transform workcell supervisors and map their states, assuming instantaneous switching faster than UC events [19, Sec. III]. State attraction bounds startup under weak invariance [34]; switched-DES control enforces mode-specific invariants under uncontrollable switches [35]. FG combines staged transfers, independent lifetimes and all-outcome finite completion.

*Requirements across adaptation.* Temporal adaptation [44], scenario-based updatability [13, 26] and Live Synthesis [12] address obligations across change. Our initialization uses scoped requirements [11] and residual reasoning [36]. Thuijsman and Reniers [39, Secs. 4–7] allow individual or simultaneous reconfiguration, requirements that depend on features, and state retention or reset. Their supervisor permits a marked continuation from every reachable state; FG requires every maximal update run to finish at a supplied controller.

*Safe adaptation paths.* Zhang et al. generate safe adaptation graphs from dependency invariants and adaptive actions, then choose minimum-cost paths [45, Secs. 3–4]. FG’s finite-handover objective has no cost optimization.

*Operational update mechanisms.* Local updating [25], PROTEOS [14], workflow updating [37], Kitsune [15] and network synthesis [27] address migration and transition safety. Ma et al. use dynamic dependencies and sufficient freeness for transaction version consistency with concurrent versions and selective blocking [23]. FG’s finite handover gives no such guarantee.

## 9 Threats to Validity

Permanently enabled UC arrivals or polling block even local commands. Threads contrasts finite UC phases with perpetual activity. Fairness-dependent waiting is outside the contract; inherited-adaptation effects on feasibility are not isolated. Decoding, history completeness and A1–A4 remain unvalidated; repair is excluded. Constructed cases assume finite startup, reporting, recovery and election; selection may favor separations, and deployed prevalence is unestablished.

Shared authorship, AI and code risk correlated faults; independent parsing covers restricted FSP. Direct-Full is in-house; Legacy’s objective differs. Pruning isolation, external solvers and randomized orders remain untested. Ratios omit unresolved costs; extended trials lack medians. Sampling can miss peaks; RSS identifies neither heap occupancy nor bottlenecks. S4 retains interruptions.

Null/LOSS results, faster Update-first, lower Eager/Update-first memory and Travel/PC limits bound gains. PC2’s merged timeout remains unresolved. Fixtures/repeats are not independent applications; timing uses one machine. Inherited inputs lack precedence; only Railcab transfers branch. Exponential separation assumes the default order.

## 10 Conclusion

FG-DUCS compiles local contracts into certified finite updates between verified controllers, preserving the selected endpoint’s continuation. Granularity changes feasibility in both directions. Evaluation shows reduced exploration and more fixed-budget decisions. These guarantees require declared requirement lifetimes, uncontrollable quiescence and faithful execution.

## Data Availability

We created and publicly released the anonymous FG-DUCS replication package for this paper [4]. The retained evidence snapshot is available at <https://github.com/research-artifact-archive/research-artifact/tree/fgducs-invariants-20261002>. The paper-matched release cited in [4], fgducs-c85-20261003, contains this revised main paper and paper/supplementary\_material.pdf: Part I is Technical Appendix A–P; Part II is Supplement S1–S5. Its reading guide, bookmarks and continuous page numbers locate the proofs, full models and experimental records cited throughout the main text. The repository README links directly to both parts and maps paper sections to evidence. The package also provides source code, all 27 application contracts, scripts and saved results; saved-evidence checks require no new performance experiments. Dependencies come from official upstream sources.

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